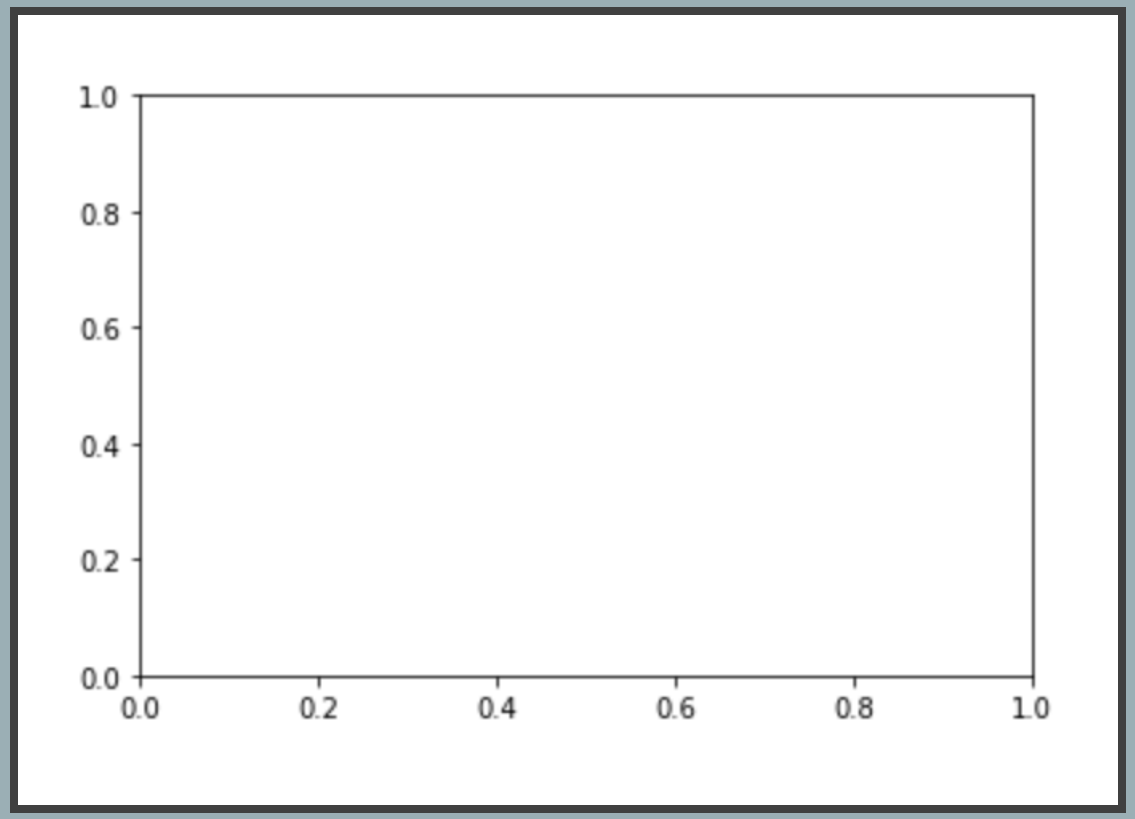
```
f = plt.figure()
```

=> Allocate space for plotting

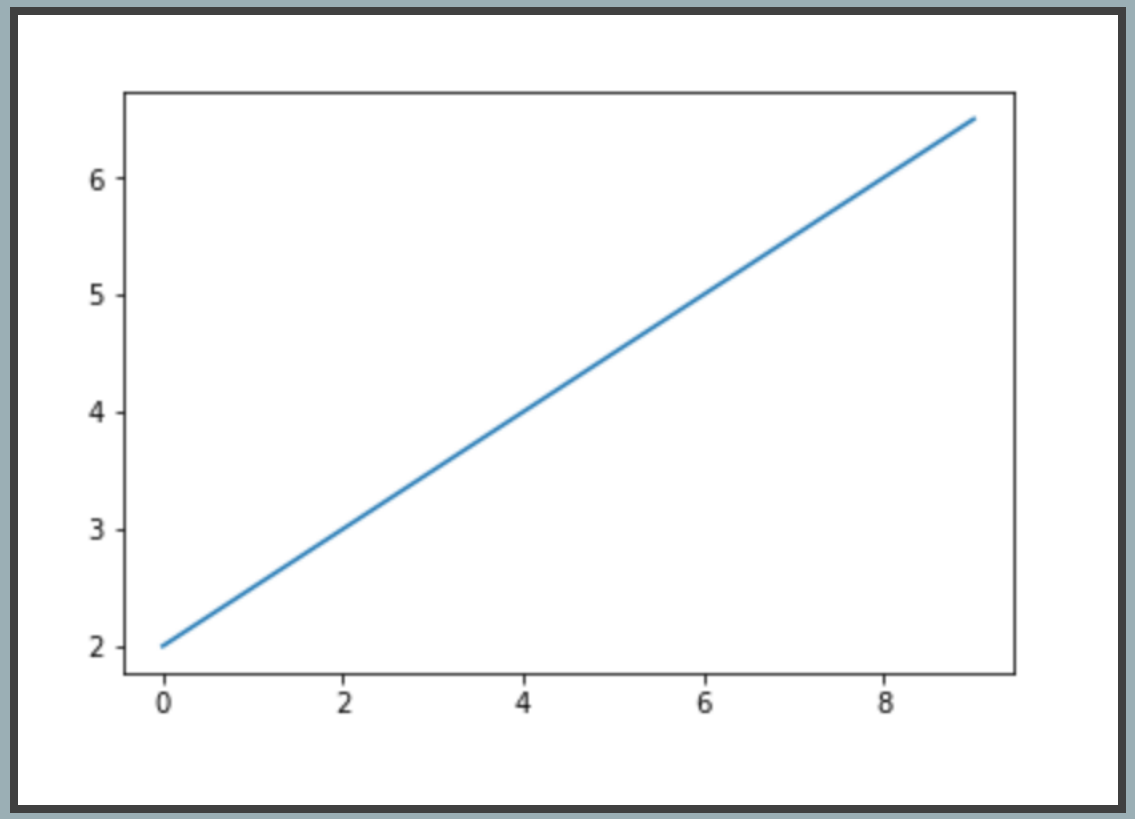


f = plt.figure()

=> Allocate space for plotting

ax = f.add_subplot(1, 1, 1)

=> Create an instance of an axes object



f = plt.figure()

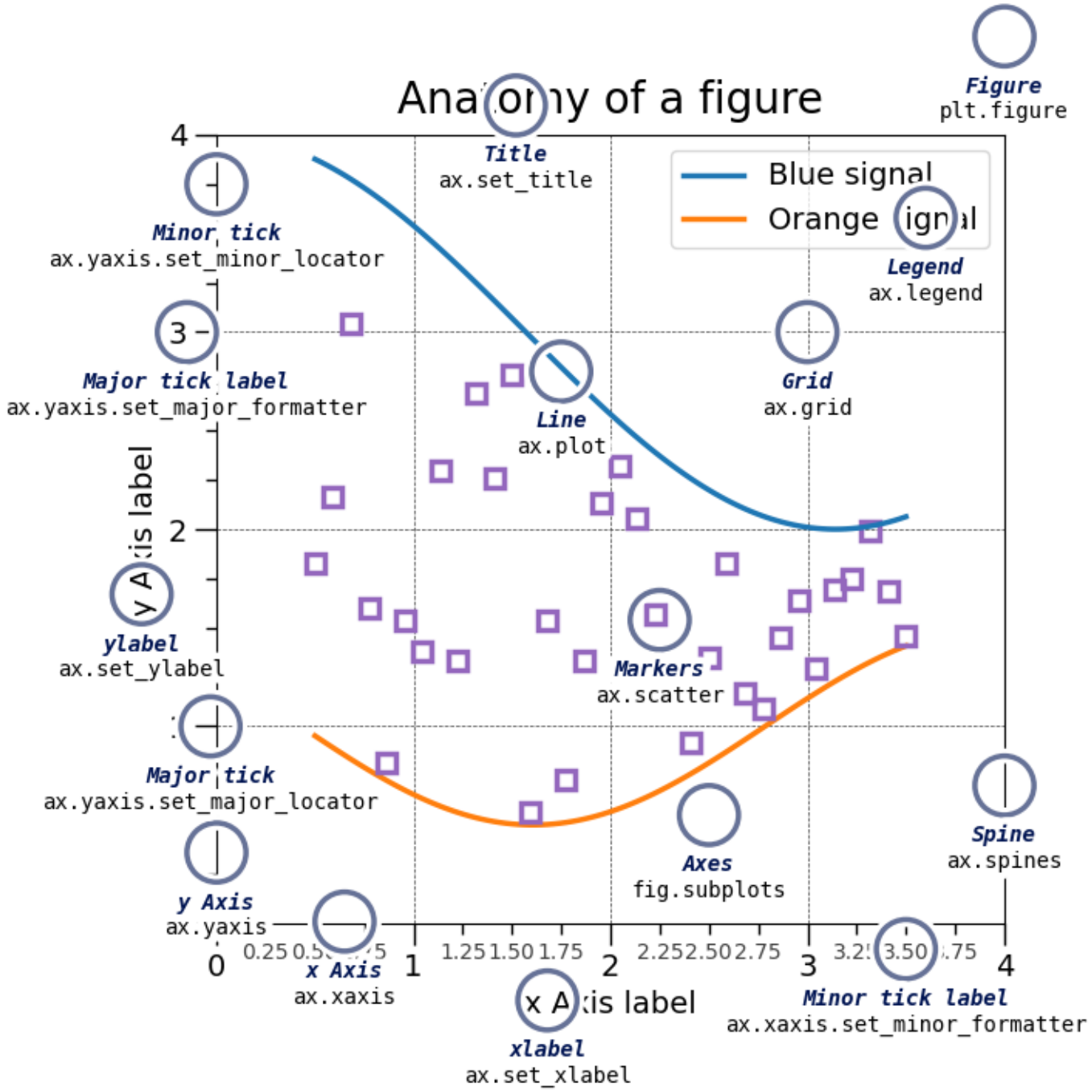
=> Allocate space for plotting

ax = f.add_subplot(1, 1, 1)

=> Create an instance of an axes object

ax.plot(x, y)

=> Plot x and y in the axes object

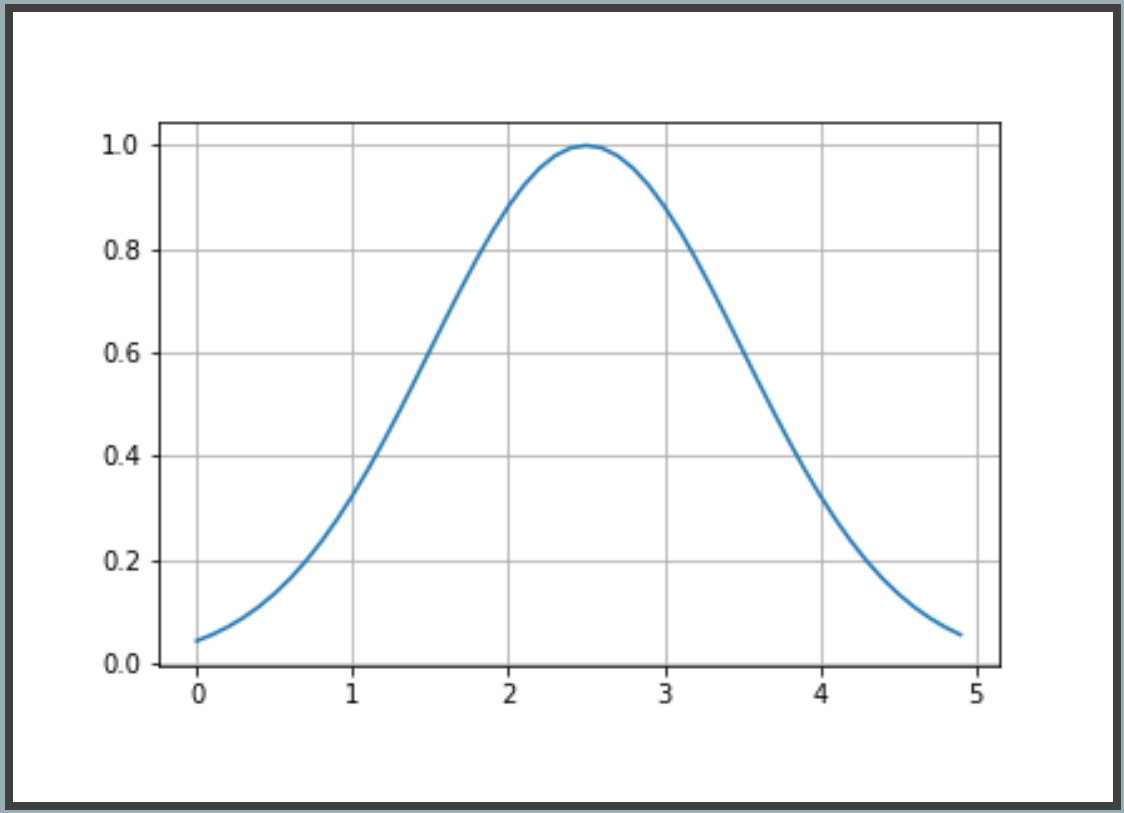


fig, ax = plt.subplots()
ax.plot(X, Y)

⇒ Command to create and return a figure (fig) AND an axes (ax) object.

np.meshgrid(): What is it?

```
def gauss1D(x, I, xc, sigx):
    arg = (x - xc) / sigx
    return np.exp( -0.5 * arg**2)
```



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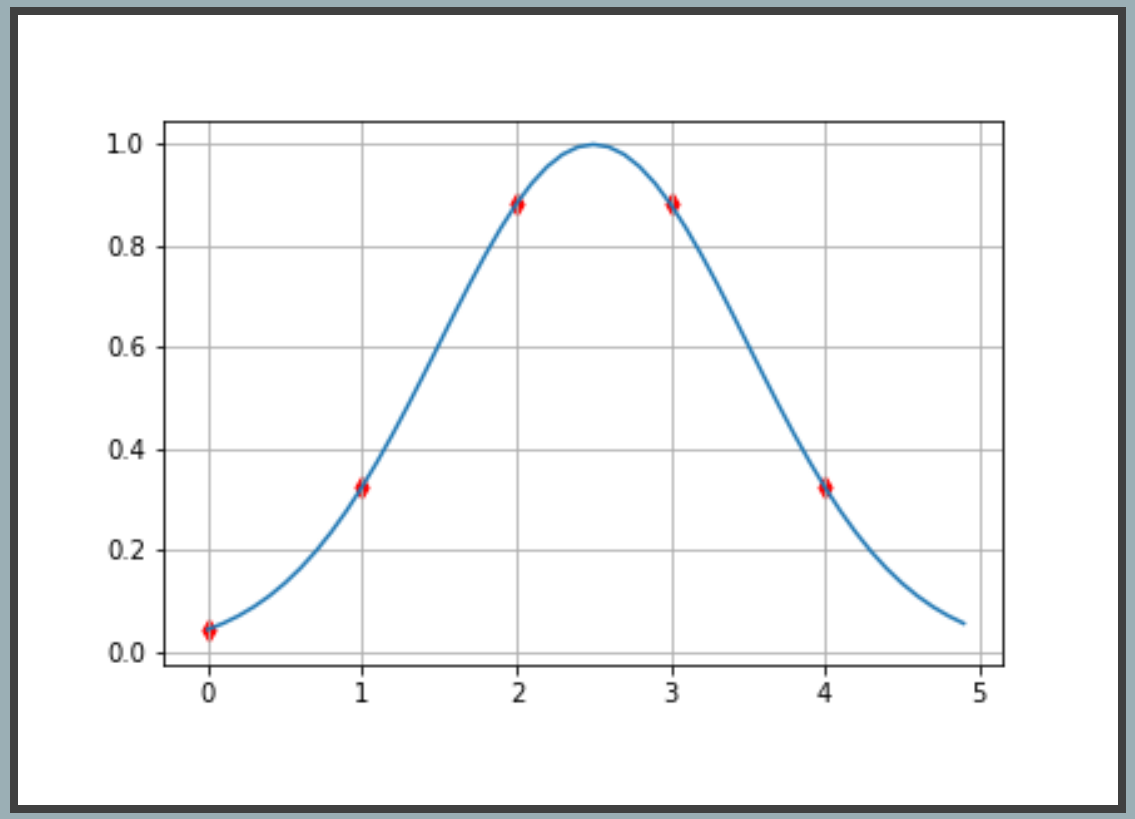


If you create a 1D array:

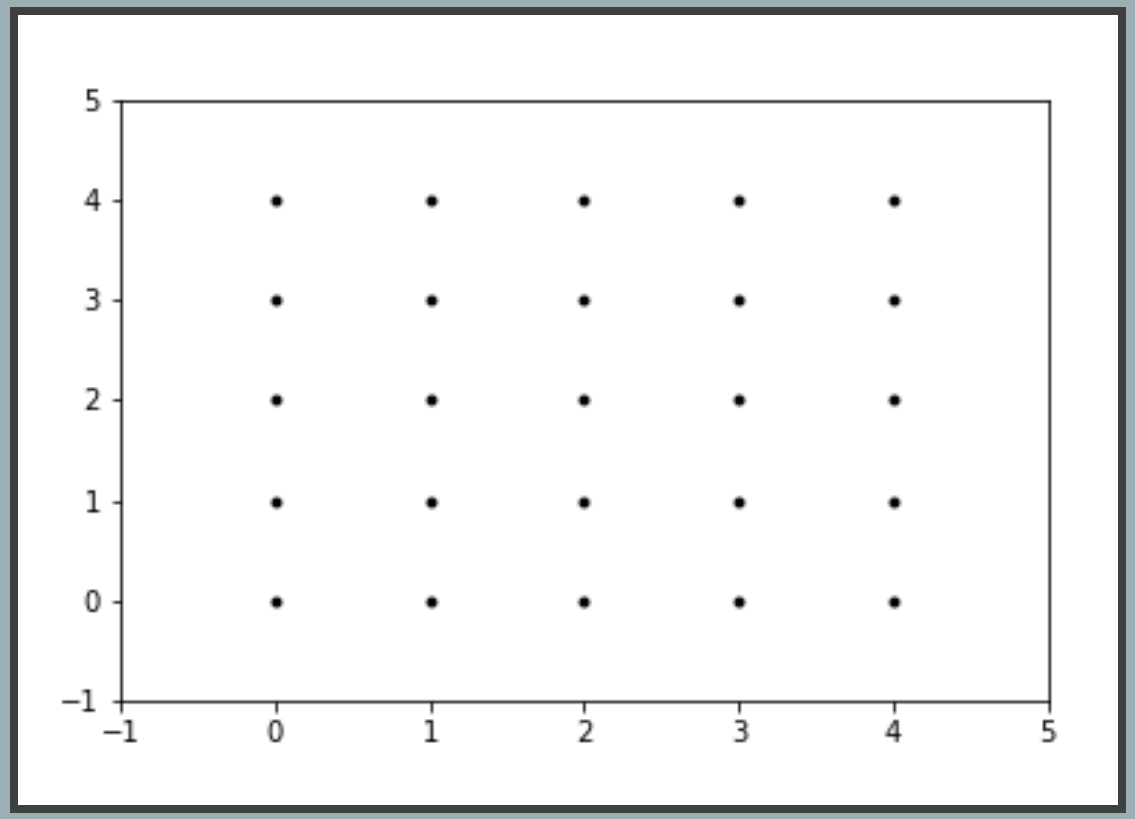
x = np.arange(0, 5, 1)

You sample your gaussian for x = 0,1…, 5

```
g1D = gauss1D(x, 1, 2.5, 1.)
ax.scatter(x, g1D, marker='d')
```

How to do with a 2D gaussian?

```
def gauss2D(x, y, I, xc, sigx, yc, sigy):
    argx = (x - xc) / sigx
        argy = (y - yc) / sigy
        g2D = np.exp(-0.5*(argx**2 + argy**2)
    return g2D
```



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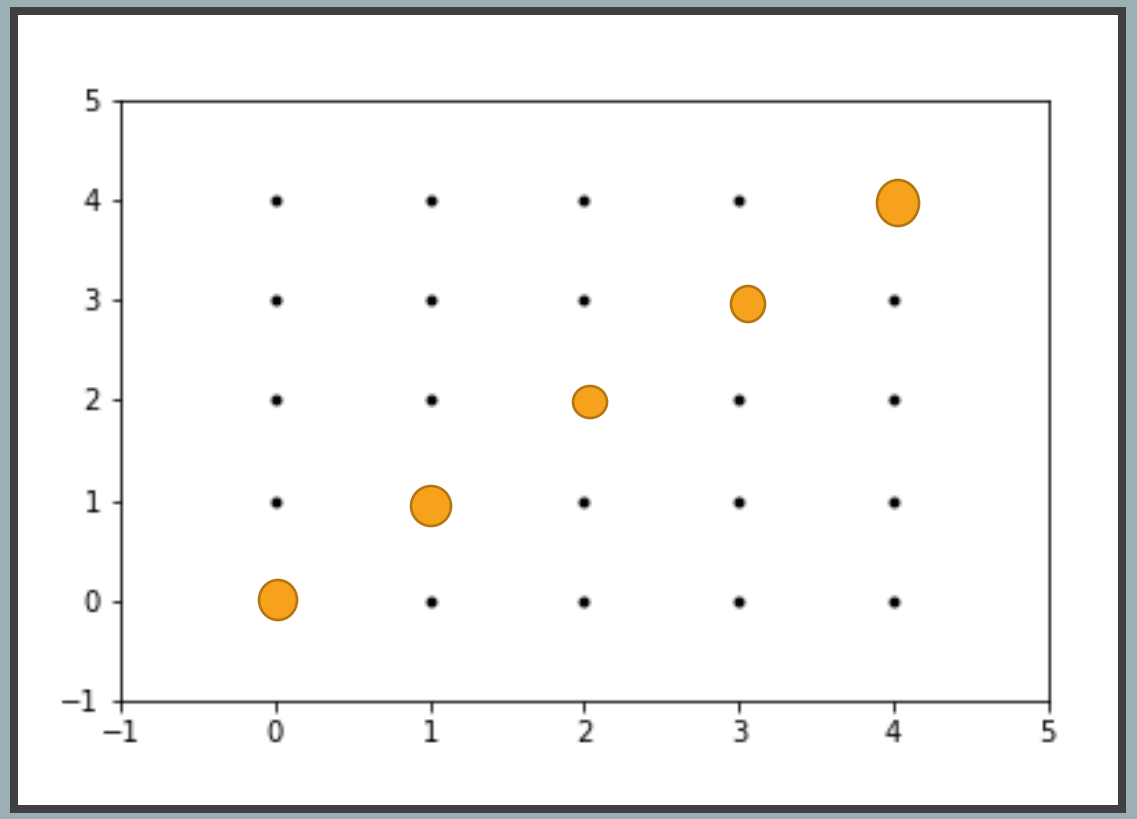


If you create two 1D array:

x = np.arange(0, 5, 1)

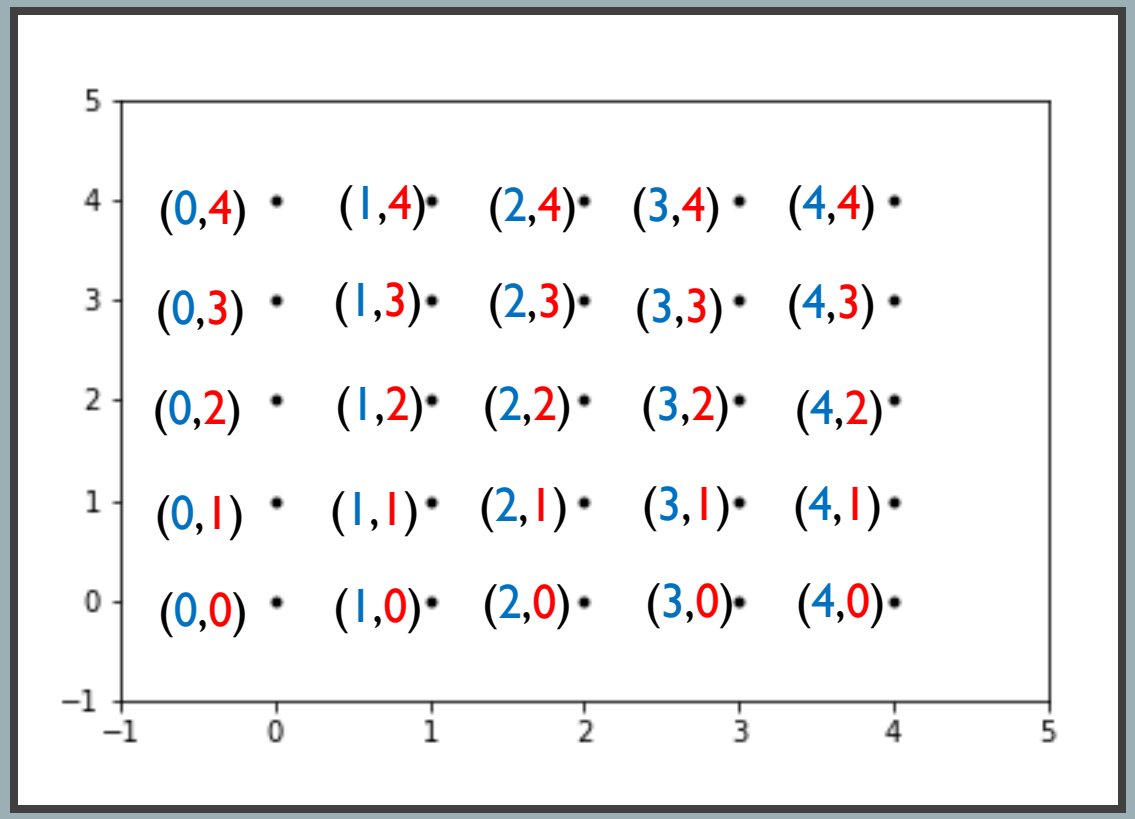
y = np.arange(0, 5, 1)

You sample your gaussian only along the diagonal [(0,0), (1,1), …, (4, 4)]

g2D = gauss2D(x, y, *args)

Meshgrid is the solution

```
def gauss2D(x, y, I, xc, sigx, yc, sigy):
    argx = (x - xc) / sigx
        argy = (y - yc) / sigy
        g2D = np.exp(-0.5*(argx**2 + argy**2)
    return g2D
```



You need a 2D array with coords. of (x,y):

```
x =   0 1 2 3 4        y =   0 0 0 0 0
      0 1 2 3 4              1 1 1 1 1
      0 1 2 3 4              2 2 2 2 2
      0 1 2 3 4              3 3 3 3 3
      0 1 2 3 4              4 4 4 4 4
```

```
X, Y = np.meshgrid(x, y)
g2D = gauss2D(X, Y, *args)
```